Post-YOLO action selection

What each method did, how it responded, and practical efficiency | October 29, 2026

Recommendation: use General Algorithmic Approach for the predefined robot plans. Use A* or UCS when the goal is an economical sequence of object reviews. The current LLM pipeline needs reliability and loading improvements before use for responsive decisions.

Recommended order for efficient, usable actions

This is an engineering recommendation for the current demo, not a universal performance score. It weighs policy fit and usable output, review-plan cost, and computation. Ties avoid overstating small timing differences. For a single review choice, probability inference can be preferred to search; if only calculation speed matters, DFS moves up.

Rank	Method	Why it is placed here	Median response
1	General robot plans	Most direct match for the demo robot policy; selects a named plan and preserves all matching action steps.	0.337 ms
2	A*	Best tier for economical review ordering; optimal on all 100 small graphs.	0.272 ms
2	UCS	Same optimal-plan result as A*; more work in the 1,000-target stress test.	0.279 ms
3	Probability enumeration	One review recommendation without graph expansion; also estimates the scene.	0.533 ms
4	Greedy best-first	Quick approximate ordering; lower compute cost but no minimum-cost guarantee.	0.279 ms
5	BFS	Complete plans, but shortest depth does not minimize weighted review effort.	0.266 ms
5	IDS	Same plan quality as BFS here, with repeated exploration and slower scaling.	0.293 ms
6	DFS	Fast plan generation; its selected review plans had the largest cost overhead.	0.267 ms
7	Qwen3-VL 2B	Rich contextual explanation, but slow and frequently rejected by current validation.	10.24 s

Observed responses on the 24 public images

Robot plans: 21 Take Action and 3 No Action. Its decisions matched 20/24 annotation-derived robot decisions (83.3%); exact plan names matched only 10/24. All 24 results agreed with the robot rules applied to the actual YOLO detections.

Each search and probability: 14 Take Action and 10 No Action. All matched the shared review rules on the actual detections in 24/24 cases; annotation-derived review agreement was 15/24 (62.5%). Identical binary decisions did not mean identical targets or review order.

LLM: 1 validated Take Action, 11 validated No Action, 11 rejected/fallback responses, and 1 no-detection bypass. The interface displayed No Action for the rejected responses, but these were **not usable decisions**. Across both repetitions, 22/46 actual inference runs fell back. After validation, first-pass agreement was 9/24 with review references and 2/24 with robot references.

Accuracy qualification: COCO labels objects, not required robot actions. Scores measure agreement with two explicit policies, not independently human-labeled action correctness. The robot reference has 21 positives and 3 negatives: always acting scores 87.5%, versus robot plans at 83.3%; balanced accuracy is 50% versus 61.9%. Review references request signal/tool review; robot references also react to people, animals, vehicles and counts. Predicted scores below 50% trigger verification, while annotations are treated as confirmed. The recommendation concerns policy fit and efficiency, not proven real-world safety.

What each algorithm actually did

Results below refer to the first measured pass over 24 distinct images.

General Algorithmic Approach - predefined robot plans

Counted eligible detections and evaluated 12 rules / 18 conditions. The first matching action plan took priority over idle plans; other matching action steps were retained. Selected 10 uncertainty, 3 vehicle, 2 people, 2 signal, 2 animal, 1 people-and-vehicles, and 1 crowd action plans; 2 sparse and 1 empty plans requested no action. Output included the plan name, matched rules and recommended steps.

A* Search - economical review sequence

Flagged tentative labels, signals and tools, then ordered all triggered reviews in batches of five using accumulated cost plus a lower-bound heuristic. First suggestions were verification in 12 images, signal checks in 2, and no action in 10. Found the minimum-cost plan on all 100 synthetic graphs. It changed review order, not the rules deciding whether review was needed.

Uniform-Cost Search - cost-first review sequence

Used the same targets and action choices as A*, expanding the lowest accumulated-cost state without a heuristic. Returned the same first-action counts as A* and optimal plans on 100/100 synthetic graphs. In the large stress case it took 215.31 ms versus A* at 193.04 ms; optimality applies within each batch, not across the complete multi-batch plan.

Probability Algorithm - strongest review recommendation

Enumerated 20 scene hypotheses and applied hand-authored review probabilities to every eligible detection. Chose the highest-probability review above its threshold: 12 verification requests, 2 signal checks and 10 no-action results. It did not search for a review route. Probability values express a teaching policy, not calibrated likelihood that an action is correct.

Greedy Best-First Search - fast approximate ordering

Expanded the state with the smallest estimated remaining review cost, ignoring cost already paid. Returned 12 verification, 2 signal-check and 10 no-action first suggestions. Plans were optimal in 31/100 synthetic graphs, with mean cost 1.10 times optimum. Its 1,000-target response was much faster than A* but did not preserve minimum-cost plans.

Breadth-First Search - shallowest complete plan

Expanded states in first-in, first-out order until every triggered target was scheduled. Returned 12 verification, 2 signal-check and 10 no-action first suggestions. All complete plans have one step per target, so a shortest-depth solution offers no weighted-cost advantage here. It was optimal in 33/100 synthetic graphs, averaging 1.08 times optimum.

Iterative Deepening Search - repeated depth-first passes

Repeated depth-limited DFS with increasing limits until it found a complete review plan. Its first suggestions and synthetic plan quality matched BFS: 12 verification, 2 signal-check, 10 no-action, and 33/100 optimal plans. Repeated exploration made it the slowest classical approach in the 1,000-target stress test, without a better action plan in these tests.

Depth-First Search - quick first complete plan

Followed a deep branch before trying alternatives. Returned 12 verification, 2 generic inspect suggestions and 10 no-action results. In the two signal cases it selected inspect rather than check_signal. It generated plans quickly, but was optimal on 0/100 synthetic graphs and averaged 1.49 times optimum. That result describes this implementation and fixture, not every possible DFS application.

Qwen3-VL 2B - visual choice plus generated explanation

Read the image and a bounded YOLO evidence list, selected environment/setting/action, then generated an explanation, with one retry when needed. The sole accepted action was to verify a tentative person detection in a skiing image. Three first-stage choices contradicted the allowed environment/setting combination; eight more images failed later explanation validation. The second pass reproduced the same decisions and statuses. Its richer output did not produce dependable action selection here.

All actions above are recommendations. This application sends no physical robot commands.

How they responded to the same scenes

Concrete recorded examples - paraphrases preserve the selected action and target.

Street intersection - COCO image 555050

YOLO returned 12 boxes. Search planned 10 reviews. The search methods agreed that review was needed, but differed in target order and effort. Costs below are the full review-plan cost, not execution time.

Method	First response	Full plan cost
A*	Check actual signal state [d4]	13.165
UCS	Check actual signal state [d4]	13.165
BFS	Check actual signal state [d2]	15.578
IDS	Check actual signal state [d2]	15.578
Greedy best-first	Check actual signal state [d5]	16.653
DFS	Inspect [d2]	17.378

Probability: check signal [d2], without planning an order. **Robot plans:** Signal-Review selected; animal, uncertainty and vehicle steps also retained. **LLM:** returned outdoor + shop + no_action; validation rejected the setting combination, so the application displayed a fallback. A* / UCS cost 13.165 versus DFS at 17.378, about 24% lower in this scene.

Three other useful examples

Scene	Observed behavior
Skiing - 43314	A* / UCS reviewed tentative skis first (cost 2.200); BFS, DFS, IDS and greedy reviewed the tentative person first (2.283). Probability and LLM also selected person verification. Robot plans requested another view for both tentative detections.
Dog - 206831	All search methods, probability and LLM returned No Action. Robot plans selected Animal-Presence and conditionally recommended pausing to review positions if moving nearby. This is a policy difference, not proof one answer is universally correct.
Boats - 503823	YOLO returned no boxes. Classical review methods and LLM returned no object-based action; the LLM was not invoked. Robot plans selected Empty-Area / No Action. None could recover the missing detections.

Scope, timing and limitations

24 COCO val2017 images: three each from eight object-based groups (signals, tools, animals, transport, kitchen/dining, home/office, sports, air/water), selected with seed 20261029 before model outputs. This is one dataset with varied scenes. Crowd annotations were excluded. YOLO26n ran once per image at 640 pixels, confidence 0.25, FP16, max 100 boxes. All methods reused those detections; only the LLM also used image pixels and could truncate the evidence list.

Hardware: Ryzen 9 5900HX and RTX 3050 Ti Laptop GPU, 4 GB. Each classical method had 120 measured calls; LLM had 48, including 2 bypasses excluded from its latency. Agreement counts distinct images from repetition 0. Timings include preparation, inference, validation and explanation; exclude YOLO, client construction, queueing, GUI and saving. First calls were separate; LLM first call was 14.75 s. Its model residency snapshot was about 3.1 GiB, with median reported loading time 6.40 s per inference. Peak memory and energy were not measured.

Synthetic tests were separate: 423 policy cases per classical method passed; 100 graphs independently checked plan cost; 1,000-target timing used three repetitions. Search cost is illustrative effort plus image-coordinate attention shifts, not physical distance. Public-image sample size is small (20/24 has about 64%-93% Wilson interval). Full latency distributions, response counts and concrete outputs accompany the PDF.

Sources: COCO data | Tested implementation: 61c9a7f. Revision changes the report only; measurements are unchanged.